

Assignment 9 : Maritime transportation

Introduction

The objective of this assignment is to become familiar with typical routing and scheduling problems in maritime transportation, and learn how these problems can be modeled.

Task 1 Extended cargo routing models

Consider the following extensions of the model (13.1)–(13.18) in Christiansen and Fagerholt (2014), (Ship Routing and Scheduling in Industrial and Tramp Shipping. Chapter 13 in P. Toth and D. Vigo (eds.), Vehicle Routing: Problems, Methods, and Applications, 2nd Edition, MOS-SIAM Series on Optimization, SIAM, Philadelphia):

1. If ship v is not used in the planning horizon, it gives a profit F_v .
2. It should be possible to fix a cargo to a particular ship.
3. Assume it might be possible to extend the time windows for start of service in a port. Assume a linear penalty cost of C^P per time unit above or below the given time windows. The lower and upper limits of the extended time windows are $\underline{T}_{iv}^P$ and $\overline{T}_{iv}^P$, respectively.

Define new notation where necessary and adjust the model (13.1)–(13.18) to take the three additional issues into account. Comment how you have considered each of the additional issues.

Task 2 Full load cargo routing model

Assume that we are operating in a segment where just one cargo can be on board a ship at any time. This type of problem is called a full load cargo routing problem. How can you adjust model (13.1)–(13.18) to take this assumption into account? Can you utilize this full load information to formulate a better model?

Task 3 A maritime inventory routing model

A maritime inventory routing model is given in (13.40)–(13.59). One problem with such a model is that the inventory levels in the end of the planning horizon usually have extreme values. Normally the optimal solution has inventory levels at it's maximum in the loading ports and at it's minimum in the unloading ports. Reformulate the model to prevent such extreme end conditions.